

ELECTROMAGNETISM

Electrostatics, Magnetodynamics, and Maxwell's Unified
Field Equations

Subject: Classical Electrodynamics & Field Matrix Foundations

Level: Core Academic Reference Notes (Undergraduate Module)

Features: Full Vector Flux Integrations & Coordinate Field Geometries Embedded

Document Status: Final Production Master File

1. ELECTROSTATICS: FIELDS AND POTENTIALS

Electrostatics addresses the properties of stationary electric charges, which establish vector force distributions within the surrounding space, known as the **Electric Field** ($\vec{E}$).

1.1 Coulomb's Law and Field Intensity

Coulomb's Law provides the basic vector interaction equation for two point charges (q_1 and q_2)

separated by a distance r in free space:

$$\vec{F} = [1 / (4\pi\epsilon_0)] \cdot (q_1 q_2 / r^2) \cdot \hat{r}$$

Where ϵ is the permittivity of free space ($8.854 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2$). The electric field intensity vector generated by a singular source charge is defined as force per unit test charge:

$$\vec{E} = \vec{F} / q_0.$$

1.2 Gauss's Law for Electrostatics

Gauss's Law links the surface integration of the normal electric field vector—formally termed the **Electric Flux** (Φ_E)—across an arbitrary closed mathematical surface to the total net enclosed localized charge matrix (q_{enclosed}):

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = q_{\text{enclosed}} / \epsilon_0$$

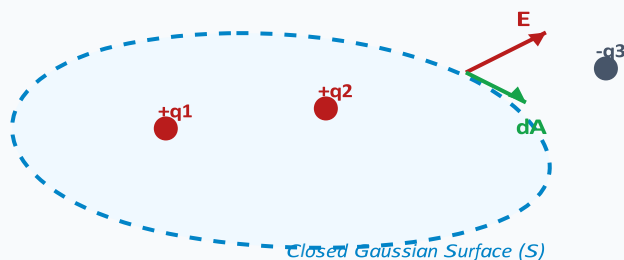


Figure 1.1: Gauss's Field Mapping Law, demonstrating how external independent metrics like $-q_3$ contribute zero net integration flux across the enclosed spatial boundary.

2. MAGNETOSTATICS: STEADY CURRENTS AND MAGNETIC FIELDS

Moving vector charge streams (steady electric currents) generate an independent non-conservative axial field distribution, defined as the **Magnetic Field** ($\vec{B}$).

2.1 The Biot-Savart Law

The Biot-Savart Law maps the differential magnetic field vector element ($d\vec{B}$) produced at a coordinate space point by an infinitesimal current element ($I d\vec{l}$):

$$d\vec{B} = [\mu_0 / (4\pi)] \cdot [I (d\vec{l} \times \vec{r}) / r^2]$$

Where μ_0 represents the permeability constant of free space ($4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$). The cross-product implies that the magnetic field is always oriented perpendicular to both the current path and the radial position vector.

2.2 Ampere's Circuital Law

Ampere's Law provides a simplified integral tool for symmetrical configurations, stating that the line integral of the magnetic field vector around any arbitrary closed loop match the total net operational current passing through the surface enclosed by the loop:

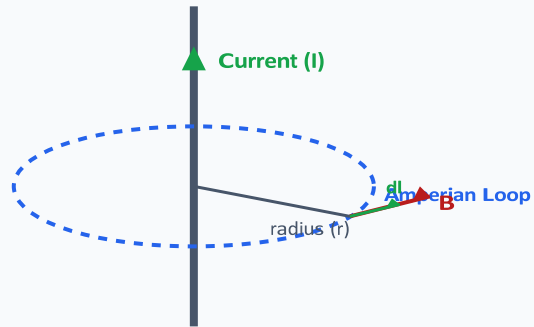


Figure 2.1: Evaluation of an Amperian Loop surrounding an infinitely straight conductor element, proving the symmetric tangential field relation $B = (\mu_o I) / (2\pi r)$.

$$\oint \vec{B} \cdot d\vec{l} = \mu_o I_{\text{enclosed}}$$

3. ELECTRODYNAMICS: INDUCTION AND TIMEVARYING FIELDS

When field metrics transition into time-varying domains, electric and magnetic fields become dynamically coupled, as detailed by Michael Faraday.

3.1 Faraday's Law of Electromagnetic Induction

Faraday discovered that changing the **Magnetic Flux** ($\Phi_B = \int \vec{B} \cdot d\vec{A}$) through a closed loop induces an Electromotive Force (EMF, denoted as $\mathcal{E}$) inside the circuit loop boundary:

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$$

The inclusion of the explicit negative sign represents **Lenz's Law**, which states that the induced current vector field will always possess an orientation that opposes the change in magnetic flux that created it (preserving conservation of energy).

4. MAXWELL'S UNIFIED ELECTROMAGNETIC FIELD MATRIX

James Clerk Maxwell unified all classical electrodynamic formulations into a cohesive system of four equations. He recognized that Ampere's Law was incomplete for time-varying fields, introducing the concept of **Displacement Current** ($\epsilon_0 \frac{d\Phi_E}{dt}$) to ensure charge conservation.

4.1 The Complete Set of Maxwell's Equations (Integral & Differential Form)

The four pillars of electrodynamics are structured in the following system:

Gauss's Law (Electric) Gauss's Law (Magnetic) Faraday's Induction Law Ampere-Maxwell Law

Axiom Name	Integral Formulation Profile	Differential Operator Profile	Physical Core Meaning
$\oint_C \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$	$\oint_C \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$		Electric fields originate from point source scalar charge densities (ρ).
$\oint_C \vec{B} \cdot d\vec{A} = 0$	$\oint_C \vec{B} \cdot d\vec{A} = 0$		Isolated magnetic monopoles do not exist; magnetic field lines form continuous closed loops.
$\oint_C \vec{E} \cdot d\vec{l} = - \frac{\partial \Phi_B}{\partial t}$	$\oint_C \vec{E} \cdot d\vec{l} = - \frac{\partial \Phi_B}{\partial t}$		A time-varying magnetic field induces a spatially curling electric field vector.
$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 (I + \epsilon_0 \frac{\partial \Phi_E}{\partial t})$	$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 (I + \epsilon_0 \frac{\partial \Phi_E}{\partial t})$		Magnetic fields curl around both moving conduction currents ($\vec{J}$) and timevarying electric fields.

5. ELECTROMAGNETIC WAVE PROPAGATION FIELDS

By decoupling Maxwell's equations in free space where source dependencies are zero ($\rho = 0, \mathbf{J} = 0$), we can apply the vector identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ to extract localized 3D spatial wave equations:

$$\begin{aligned}\nabla^2 \vec{E} &= \mu_0 \epsilon_0 (\partial^2 \vec{E} / \partial t^2) \\ \nabla^2 \vec{B} &= \mu_0 \epsilon_0 (\partial^2 \vec{B} / \partial t^2)\end{aligned}$$

This mathematical derivation proves the existence of self-propagating electromagnetic waves that travel at a speed determined exclusively by free-space constants:

$$c = 1 / \sqrt{(\mu_0 \epsilon_0)} \approx 3 \times 10^8 \text{ m/s}$$

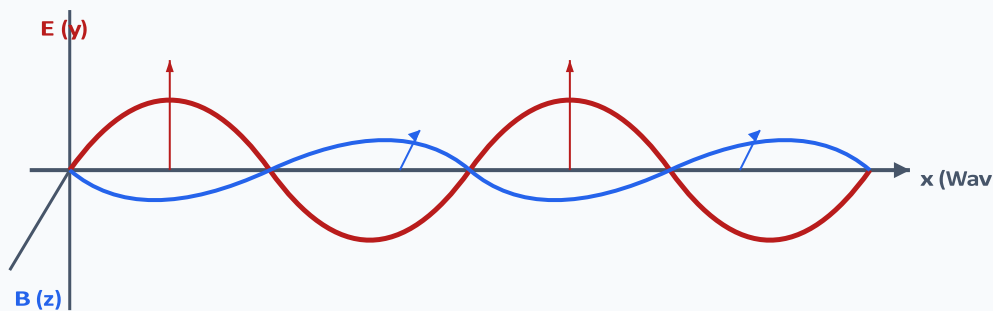


Figure 5.1: 3D Orthogonal Vector Mapping of a plane electromagnetic wave, illustrating how the electric field ($\mathbf{E}$), magnetic field ($\mathbf{B}$), and propagation wave vector ($\mathbf{k}$) are mutually perpendicular.